

Chapter 4

Attributes-based Re-Identification

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Abstract Automated person re-identification using only visual information from public-space CCTV video is challenging for many reasons, such as poor resolution or challenges involved in dealing with camera calibration. More critical still, the majority of clothing worn in public spaces tends to be non-discriminative and therefore of limited disambiguation value. Most re-identification techniques developed so far have relied on low-level visual-feature matching approaches that aim to return matching gallery detections earlier in the ranked list of results. However, for many applications an initial probe image may not be available, or a low-level feature representation may not be sufficiently invariant to viewing condition changes as well as being discriminative for re-identification. In this chapter, we show how mid-level “semantic attributes” can be computed for person description. We further show how this attribute-based description can be used in synergy with low-level feature descriptions to improve re-identification accuracy when an attribute-centric distance measure is employed. Moreover, we discuss a “zero-shot” scenario in which a visual probe is unavailable but re-identification can still be performed with user-provided semantic attribute description.

4.1 Introduction

Person re-identification, or *inter-camera entity association*, is the task of recognising an individual in diverse scenes obtained from non-overlapping cameras. In particular, for surveillance applications performed over space and time, an individual

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disappearing from one view would need to be differentiated from numerous possible targets and matched in one or more other views at different locations and times. Potentially each view may be taken from a different angle, featuring different static and dynamic lighting conditions, degrees of occlusion and other view-specific variables.

Relying on manual re-identification in large camera networks is prohibitively costly and inaccurate. Operators are often assigned more cameras to monitor than is optimal and manual matching can be prone to attentive gaps [19]. Moreover, baseline human performance is determined by individual operator’s experience amongst other factors. It is difficult to transfer this expertise directly between operators without knowledge being affected by operator-bias [45].

As public space camera networks have grown quickly in recent years, there has also been an increasing interest in the computer vision community for developing automated re-identification solutions. These efforts have primarily focused on two strategies: (i) developing feature representations which are discriminative for identity, yet invariant to view angle and lighting [12, 37, 3]; and (ii) learning methods to discriminatively optimise parameters of a re-identification model [50]. Until now, automated re-identification remains largely an unsolved problem due to the underlying challenge that most visual features are either insufficiently discriminative for cross-view entity association, especially with low resolution images, or insufficiently robust to viewing condition changes.

In this chapter, we take inspiration from the operating procedures of human experts [33, 43, 8] and recent research in attribute learning for classification [21] in order to introduce a new mid-level *semantic attribute* representation.

When performing person re-identification, human experts rely upon matching appearance or functional attributes that are discrete and unambiguous in interpretation, such as hair-style, shoe-type or clothing-style [33]. This is in contrast to the continuous and more ambiguous quantities measured by contemporary computer vision based re-identification approaches using visual features such as colour and texture [12, 37, 3]. This attribute-centric representation is similar to a description provided verbally to a human operator, e.g. by an eye-witness. We call this task attribute-profile identification, or *zero-shot re-identification*. Furthermore, we will show in our study that humans and computers have important differences in attribute-centric re-identification. In particular descriptive attributes that are favoured by humans may not be the most *useful* or *computable* for fully automated re-identification because of variance in the ability of computer vision techniques to detect each attribute and variability in how discriminative each attribute is across the entire population.

This approach of measuring similarity between attributes rather than within the feature-space has two advantages: (i) it allows re-identification (from a probe image) and identification (from a verbal description) to be performed in the same representational space; and (ii) as attributes provide a very different type of information to low-level features, which can be considered a separate modality, they can be fused together with low-level features to provide more accurate and robust re-identification.

4.2 Problem Definitions

4.2.1 The Re-Identification Problem

Contemporary approaches to re-identification typically exploit low-level features (LLFs) such as colour [29], texture, spatial structure [3], or combinations thereof [13, 37, 4], because they can be relatively easily and reliably measured, and provide a reasonable level of inter-person discrimination together with inter-camera invariance.

Once a suitable representation has been obtained, nearest-neighbour [3] or model-based matching algorithms such as support-vector ranking [37] may be used for re-identification. In each case, a distance metric (e.g. Euclidean or Bhattacharyya) must be chosen to measure the similarity between two samples. There is now a body of work on discriminatively optimising re-identification models or distance metrics [15, 47, 2, 50] as well as discriminatively learning the low-level features themselves [24]. Other complementary aspects of the re-identification problem have also been pursued to improve performance, such as improving robustness by combining multiple frames worth of features along a trajectory tracklet [4], between sets [49], in a group [46], and learning the topology of camera networks by learning inter-camera activity correlations [27] in order to reduce matching search space and hence reduce false-positives.

4.2.2 Attributes as Representation

Attribute based modelling has recently been exploited to good effect in object [21] and action [26, 11] recognition. To put this in context: in contrast to low-level features or high-level classes or identities, attributes provide the mid-level *description* of both classes and instances. There are various unsupervised (e.g. PCA or topic-models) or supervised (e.g. neural networks) modelling approaches which produce data-driven mid-level representations. These techniques aim to project the data onto a basis set defined by the assumptions of the particular model (e.g. maximisation of variance, likelihood, or sparsity). In contrast, attribute learning focuses on representing data instances by projecting them onto a basis set defined by domain-specific axes which are semantically meaningful to humans. Recent work in this area has also examined the exploitation of the constantly growing semantic web in order to automatically retrieve visual data correlating to relevant metatext [10] and vice-versa for visual retrieval using metatext queries [38].

Semantic attribute representations have various benefits: (i) In re-identification, a single pair of images may be available for each target – which can be seen as a challenging case of “one-shot” learning. In this case attributes can be more powerful than low-level features [21, 41, 26] because they provide a form of transfer learning as attributes are learned from a larger dataset *a priori*; (ii) they can be used

synergistically in conjunction with raw data for greater effectiveness [26]; and (iii) they are a suitable representation for direct human interaction, therefore allowing searches to be specified, initialised or constrained using human-labelled attribute-profiles [21, 41, 20].

4.2.3 *Attributes for Identification*

One view of attributes is as a type of transferable context [48] in that they provide auxiliary information about an instance to aid in (re)-identification. Here they are related to the study of soft-biometrics, which aims to enhance biometric identification performance with ancillary information [18, 9]. High-level features such as ethnicity, gender, age or indeed identity itself would be the most useful to us for re-identification. However, soft biometrics are exceptionally difficult to reliably compute in typical surveillance video as visual information is often impoverished and individuals are often at “stand-off distances” as well as in unconstrained or unknown viewing angles.

Alternatively attributes can be used for semantic attribute-profile identification (c.f. zero-shot learning [21]), in which early research has aimed to retrieve people matching a verbal attribute description from a camera network [43]. However, this has only been illustrated on relatively simple data with a small set of similarly-reliable facial attributes. We will illustrate in this study that one of the central issues for exploiting attributes for general automated (re)-identification is dealing with their unequal and variable informativeness and reliability of measurement from raw imagery data.

In this chapter, we move towards leveraging semantic mid-level attributes for automated person identification and re-identification. Specifically, we make four main contributions as follows. In Section 4.3.1 we introduce an ontology of attributes based on a subset from a human expert defined larger set [33]. These were selected for being relatively more reliable to compute whilst also discriminative for identification in typical populations. We evaluate our ontology from the perspective of both human-centric and automation-centric purposes and discuss considerations for successful ontology selection. In Section 4.3.6 we show how to learn an attribute-space distance metric to optimally weight attributes for re-identification, and do so in a synergistic way with low-level features. We evaluate our model in Section 4.4 and show significantly improved re-identification performance compared to conventional feature-based techniques on the two largest benchmark datasets. In the subsequent sections, we provide additional analysis and insight into the results, including contrast against zero-shot re-identification from attribute-profile descriptions.

4.3 Computing Attributes for Re-Identification

4.3.1 Ontology Selection

The majority of recent work on attributes looks to human expertise in answer to the question as to which attributes to learn. Typically, ontology selection is performed manually prior to research or via learning from existing metadata [5]. Hand-picked ontologies can be broadly categorised as top-down and bottom-up. In the top-down case, ontology selection may be predicated on the knowledge of experienced human domain-experts. In the latter it may be based on the intuition of vision researchers, based on factors such as how detectable an attribute might be with available methods or data availability.

For the purposes of automated re-identification, we are concerned with descriptions that permit us reliably discriminate; that is to say we wish to eliminate identity ambiguity between individuals. Ontology selection therefore is guided by two factors: *computability* and *usefulness*. That is, *detectable* attributes, which can be detected reliably using current machine learning methods and available data [11], and *discriminative* (informative) attributes which, if known, would allow people to be effectively disambiguated [28].

The notion of discriminative attributes encompasses a nuance. Humans share a vast prior pool of potential attributes and experience. If required to describe a person in a way which uniquely identifies them against a gallery of alternatives, they typically choose a short description in terms of the rare attributes which uniquely discriminate the target individual (e.g. imperial moustache). In contrast, in the ideal discriminative ontology of attributes for automated processing, each attribute should be uncorrelated with all others, and should occur in exactly half of the population (e.g. male versus female). In this way no one attribute can distinguish a person uniquely, but together they effectively disambiguate the population: a “binary search” strategy. There are two reasons for this: constraining the ontology size, and training data requirement.

Ontology size: Given a “binary search” ontology, any individual can be uniquely identified among a population of n candidates with only an $O(\log(n))$ sized attribute ontology or description. In contrast, the single rare-attribute strategy favoured by people means that while a person may be identified with a short length 1 attribute description, an ontology size and computation size $O(n)$ may be required to describe, interpret and identify this person.

Training data: Given a “binary search” ontology, each training image may be reused and be (equally) informative for all n attributes (attributes are typically positive for half the images). In contrast, the single rare-attribute strategy would require an infeasible n times as much training data, because different data would be needed for each attribute (e.g. finding a significant number of wearers of imperial moustaches) to train the detectors). In practice, rare attributes do not have enough training data to learn good classifiers, and are thus not reliably *detectable*. A final consideration is the visual subtlety of the attributes, which humans may be able to easily pick out



Fig. 4.1: Positive instances of our ontology from (top) the VIPeR and (bottom) the PRID datasets.

Table 4.1: Our attribute ontology for re-identification.

redshirt	blueshirt	lightshirt
darkshirt	greenshirt	nocoats
notlightdarkjeanscolour	darkbottoms	lightbottoms
hassatchel	barelegs	shorts
jeans	male	skirt
patterned	midhair	darkhair
bald	hashandbagcarrierbag	hasbackpack

based on their lifetime of experience but which would require prohibitive amounts of training data as well as feature/classifier engineering for machines to detect.

Whether or not a particular ontology is detectable and discriminative cannot therefore be evaluated prior to examination of representative data. However, given a putative ontology and a representative and annotated training set, the detectability of the ontology can be measured by the test performance of the trained detectors whilst the discriminativeness of the ontology can be measured by the mutual information (MI) between the attributes and person identity. The question of how to trade off discriminativeness and detectability when selecting an ontology on the basis of maximum predicted performance is not completely clear [22, 23]. However, we will take some steps to address this issue in Section 4.3.6.

4.3.2 Ontology Creation and Data Annotation

Given the considerations discussed in the previous section, we select our ontology jointly based on four criteria. (i) We are informed by the operational procedures of human experts [33] as well as (ii) prioritising suitable findings from [44, 22, 23, 38], (iii) whether the ontology is favourably distributed in the data (binary search) and (iv) those which are likely to be detectable (sufficient training data and avoiding subtlety).

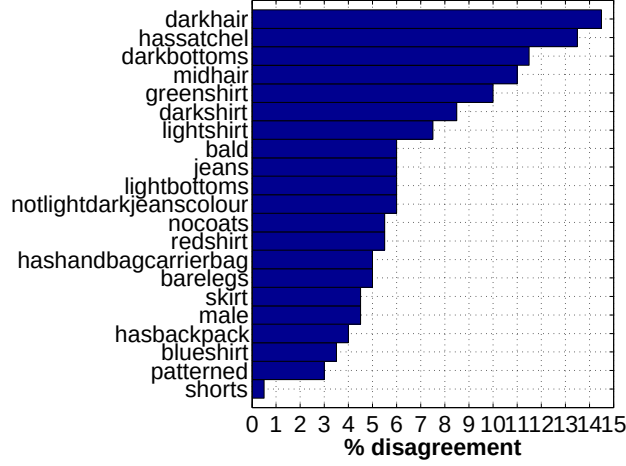


Fig. 4.2: Annotation disagreement error frequencies for two annotators on PRID.

Specifically, we define the following space of $N_a = 21$ binary attributes (Table 4.1). Ten of these attributes are related to colour, one to texture and the remaining ten are related to soft biometrics. Fig. 4.1 shows a visual example of each attribute¹.

Human annotation of attribute labels is costly in terms of both time and human effort. Due to the semantic nature of the attributes, accurate labelling can be especially challenging for cases where data are visually impoverished. Typically problems can arise where (i) ontology definition allows for ambiguity between members of the ontology, and (ii) boundary cases are difficult for an annotator to binarily classify with confidence. These circumstances can be natural places for subjective labelling errors [42].

To investigate the significance of this issue, we independently double-annotated the PRID dataset [15] for our attribute ontology. Fig. 4.2 illustrates frequency of label disagreements for each attribute in the PRID dataset measured as the Hamming distance between all annotations for that attribute across the dataset:

For attributes such as *shorts* or *gender*, uncertainty and therefore error is low. However, attributes whose boundary cases may be less well globally agreed upon can be considered to have the highest relative error between annotators. For example, in Fig. 4.2 attributes *hassatchel* and *darkhair* are most disagreed upon since lighting variations make determining darkness of hair difficult in some instances and satchel refers to a wide variety of rigid or non-rigid containers held in multiple ways. This means that attributes such as *darkhair* and *hassatchel* may effectively be subject to a significant rate of label noise [51] in the training data and hence perform poorly. This adds another source of variability in reliability of attribute detection which will have to be accounted for later. Fig. 4.3 illustrates pairs of individuals in the PRID dataset whose shared attribute-profiles were the most disagreed upon.

¹ We provide our annotations here: [http://www.eecs.qmul.ac.uk/\\$\sim\\$sim\\$rlayne/](http://www.eecs.qmul.ac.uk/$\sim$sim$rlayne/)

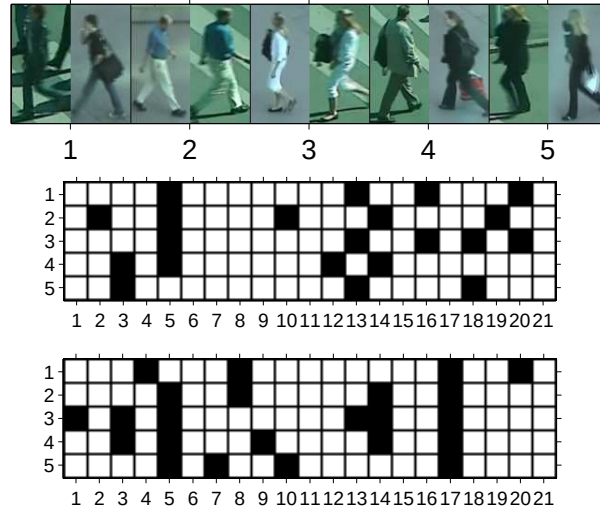


Fig. 4.3: Top 5 pairs of pedestrian detections in PRID where annotators disagreed most (top row). Annotator #1's labels (middle), Annotator #2's labels (bottom). Each row is an attribute-profile for a pair of detections, columns are attributes and are arranged in the same order as Fig 4.2.

The figure highlights the extent of noise that can be introduced through semantic labelling errors, a topic we will revisit later in Section 4.3.6.

4.3.3 Feature Extraction

To detect attributes, we first select well-defined and informative low-level features with which to train robust classifiers. We wish to choose a feature which is also typically used for re-identification in order to enable later direct comparison between conventional and attribute-space re-identification in a way which controls for the input feature used. Typical descriptors used for re-identification include the Symmetry Driven Accumulation of Local Features (SDALF) [3] and Ensemble of Localised Features (ELF) [13].

The content of our ontology includes semantic attributes such as jeans, shirt colours, gender. We can infer that the information necessary for humans to distinguish these items is present visually, and wish to select a feature that incorporates information pertaining to colour, texture and spatial information. For our purposes, SDALF fulfils the requirements for our ontology but does not produce positive semi-definite distances, therefore ruling it out for classification using kernel methods. As a result, we therefore exploit ELF.

To that end, we first extract an 2784-dimensional low-level colour and texture feature vector denoted $\mathbf{x}$ from each person image I following the method in [37]. This consists of 464-dimensional feature vectors extracted from six equal sized horizontal strips from the image. Each strip uses 8 colour channels (RGB, HSV and YCbCr) and 21 texture filters (Gabor, Schmid) derived from the luminance channel. We use the same parameter choices for γ , λ , θ and σ^2 as proposed in [37] for Gabor filter extraction, and for τ and σ for Schmid extraction. Finally, we use a bin size of 16 to quantise each channel.

4.3.4 Attribute Detection

4.3.4.1 Classifier training and attribute feature construction

We train Support Vector Machines (SVM) [40] to detect attributes. We use Chang et al.'s LIBSVM [6] and investigate Linear, RBF, χ^2 and Intersection kernels. We select the intersection kernel as it compares closely with χ^2 but is faster to compute².

For each attribute, we perform cross validation to select values for SVM slack parameter C from the set $C \in \{-10, \dots, 10\}$ with increments of $\varepsilon = 1$. The SVM scores are probability mapped, so each attribute detector i outputs a posterior $p(a_i|\mathbf{x})$. We follow the standard approach for mapping SVM scores to posterior probabilities [36] as implemented by libSVM [6].

4.3.4.2 Spatial Feature Selection

Since some attributes (e.g. shorts) are highly unlikely to appear outside of their expected spatial location, one might ask whether it is possible to improve performance by discriminatively selecting or weighting the individual strips within the feature vector (Section 4.3.3). We experimented with defining a kernel for each strip as well as for the entire image, and training multi-kernel learning SVM using the DOGMA library with *Obscure* as in classifiers [34, 35]. This approach discriminatively optimises the weights for each kernel in order to improve classifier performance and has been shown to improve performance when combining multiple features. However in this case it did not reliably improve on the conventional SVM approach, presumably due to the relatively sparse and imbalanced training data being insufficient to correctly tune the inter-kernel weights.

² Our experiments on LIBSVM performance vs. attribute training time show the intersection kernel as being a good combination of calculation time and accuracy. For example, training the attribute ontology results in 65.4% mean accuracy with 0.8 hours training for the intersection kernel, as compared to the χ^2 kernel (63.8% with 4.1 hours), the RBF kernel (65.9% with 0.76 hours and the linear kernel (61.8% with 1.2 hours) respectively with LIBSVM. Although RBF is computed slightly faster and has similar accuracy, we select the intersection kernel overall, since the RBF kernel would require cross-validating over a second parameter. Providing LibSVM with pre-built kernels reduces training time considerably in all cases.

4.3.4.3 Imbalanced Attribute Training

The prevalence of each attribute in a given dataset tends to vary dramatically and some attributes have a limited number of positive examples in an absolute sense as a result. This imbalance can cause discriminative classifiers such as SVMs to produce biased or degenerate results. There are various popular approaches to dealing with imbalanced data [14], such as synthesising further examples from the minority class to improve the definition of the decision boundary, for example using SMOTE [7] or weighting SVM instances or mis-classification penalties [14, 1]. However, neither of these methods outperformed simple subsampling in our case.

To avoid bias due to imbalanced data, we therefore simply train each attribute detector with all the positive training examples of that attribute, and obtain the same number of negative examples by sub-sampling the rest of the data at regular intervals.

4.3.4.4 Mid-level Attribute Representation

Given the learned bank of attribute detectors, at test time we generate mid-level features as $1 \times N_a$ sized vectors of classification posteriors which we use to represent the probability that each attribute is present in the detection. Effectively we have projected the high dimensional, low-level features onto a mid-level, low-dimensional semantic attribute space. In particular, each person image is now represented in semantic attribute space by stacking the posteriors from each attribute detector into the N_a dimensional vector: $A(\mathbf{x}) = [p(a_1|\mathbf{x}), \dots, p(a_{N_a}|\mathbf{x})]^T$.

4.3.5 Attribute Fusion with Low-level Features

To use our attributes for re-identification, we can define a distance solely on the attribute space, or use the attribute distance in conjunction with conventional distance between low-level features such as SDALF [3] and ELF [12]. SDALF provides state of the art performance for a non-learning nearest-neighbour (NN) approach while ELF has been widely used by model-based learning approaches [37, 46]. We also use it as the feature for our attribute detectors in Section 4.3.3.

We therefore introduce a rather general formulation of a distance metric between two images I_p and I_g which combines both multiple attributes and multiple low-level features as follows:

$$d_{\mathbf{w}^L, \mathbf{w}^A}(I_p, I_g) = \sum_{l \in LL} w_l^L d_l^L(L_l(I_p), L_l(I_g)) + d_{\mathbf{w}^A}^A(A(I_p), A(I_g)). \quad (4.1)$$

Here Eq. (4.1) (first term) corresponds to the contribution from a set LL of low-level distance measures, where $L_l(I_p)$ denotes extraction of type l low-level features from

image I_p , d_l^L denotes the distance metric defined for low-level feature type l , and w_l^L is a weighting factor for each feature type l . Eq. (4.1) (second term) corresponds to the contribution from our attribute-based distance metrics. Where $A(I_p)$ denotes the attribute encoding of image I_p . For the attribute-space distance we experiment with two metrics: weighted L1 (Eq. 4.2) and weighted Euclidean (Eq. 4.3).

$$d_{\mathbf{w}^A}^A(I_p, I_g) = (\mathbf{w}^A)^T |(A(\mathbf{x}_p) - A(\mathbf{x}_g))|, \quad (4.2)$$

$$d_{\mathbf{w}^A}^A(I_p, I_g) = \sqrt{\sum_i w_i^A (p(a_i|\mathbf{x}_p) - p(a_i|\mathbf{x}_g))^2}. \quad (4.3)$$

4.3.6 Attribute Selection and Weighting

As discussed earlier, all attributes are not equal due to variability in how reliably they are measured due to imbalance, subtlety (detectability) and how informative they are about identity (discriminability). How to account for variable detectability and discriminability of each attribute ($\mathbf{w}^A$), and how to weight attributes relative to low-level features ($\mathbf{w}^{LL}$) are important challenges which we discuss now.

Exhaustively searching the N_a dimensional space of weights directly to determine attribute selection and weighting is computationally intractable. However, we can re-formulate the re-identification task as an optimisation problem and apply standard optimisation methods [32] to search for a good configuration of weights.

Importantly, we only search $|\mathbf{w}^A| = N_a = 21$ parameters for the within-attribute-space metric $d_{\mathbf{w}^A}^A(\cdot, \cdot)$, and one or two parameters for weighting attributes relative to low-level features. In contrast to previous learners for low-level features [37, 47, 50] which must optimise hundreds or thousands of parameters, this gives us considerable flexibility in terms of computation requirement of the objective.

An interesting question is therefore what is the ideal criterion for optimisation. Previous studies have considered optimising, e.g. relative rank [37] and relative distance [50, 15]. While effective, these metrics are indirect proxies for what the re-identification application ultimately cares about, which is the average rank of the true match to a probe within the gallery set, which we call Expected Rank (ER). That is, how far does the operator have to look down the list before finding the target. See Section 4.4 for more discussion.

We introduce the following objective for expected rank:

$$ER = \frac{1}{|P|} \sum_{p \in P} \sum_{g \in G} L_{\mathbf{w}}(D_{pp}, D_{pg}) + \lambda \|\mathbf{w} - \mathbf{w}_0\|, \quad (4.4)$$

where D_{pg} is the matrix of distances (Eqs. (4.1)) from probe image p to gallery image g ; L is a loss function which can penalise the objective according to the relative distance of the true match D_{pp} versus false matches D_{pg} ; and $\mathbf{w}_0$ is a regulariser

Algorithm 1 Attributes-based Re-identification

Training
for all Attribute **do**
 Subsample majority class to length of minority class
 Cross-validate to obtain parameter C that gives best average accuracy.
 Retrain SVM on all training data with selected C
end for
Determine inter and intra-attribute weighting $\mathbf{w}$ by minimising Eq. (4.4).

Testing (Re-identification)
for all Person $\mathbf{x}_g \in$ gallery set **do**
 Classify each attribute a
 Stack attribute posteriors into person signature $A(\mathbf{x}_g)$.
end for
for all Person $\mathbf{x}_p \in$ probe set **do**
 Classify each attribute a
 Stack attribute posteriors into person signature $A(\mathbf{x}_p)$.
 Compute distance to gallery set fusing attribute and LLF cues with weight $\mathbf{w}$. (Eq. (4.1))
 Nearest-neighbour re-identification in gallery according to their similarity to person $\mathbf{x}_p$.
end for

bias with strength λ . To complete the definition of the objective, we define the loss function L as in Eq. 4.5. That is, imposing a penalty every time a false match is ranked ahead of the true match. ($\mathbf{I}$ is an indicator function which returns 1 when the parameter is true.) The overall objective (Eq. (4.4)) thus returns the expected rank of the true match. This is now a good objective, because it directly reflects the relevant end-user metric for effectiveness of the system. However it is hard to efficiently optimise because it is non-smooth: a small change to the weights $\mathbf{w}$ may have exactly zero change to the expected rank (the optimisation surface is piece-wise linear). We therefore soften this loss-function using a sigmoid, as in Eq. (4.6), which is now smooth and differentiable. This finally allows efficient gradient-based optimisation with Newton [25] or conjugate-gradient methods [32].

$$L_{\mathbf{w}}^{HardRank,ER} = \mathbf{I}(d_{pp} - d_{pg} > 0). \quad (4.5)$$

$$L_{\mathbf{w}}^{Sigmoid,ER} = \sigma(d_{pp} - d_{pg}). \quad (4.6)$$

We initialise $\mathbf{w}_{initial} = 1$. To prevent over fitting, we use regularisation parameters $\mathbf{w}_0=1$, and $\lambda = 0.2$ (i.e., everything is assumed to be equal *a priori*) and set the sigmoid scale to $k = 32$. Finally for fusion with low-level features (Eq. 4.1), we use both SDALF and ELF.

In summary, this process uses gradient-descent to search for a setting of weights $\mathbf{w}$ for each LLF and for each attribute (Eq. (4.1)) that will (locally) minimise the expected rank within the gallery of the true match to each probe image (Eq. (4.4)). See Algorithm 1 for an overview of our complete system.

4.4 Experiments

4.4.1 Datasets

We select two challenging datasets with which to validate our model, VIPeR [12] and PRID [15]. VIPeR contains 632 pedestrian image pairs from two cameras with different viewpoint, pose and lighting. Images are scaled to 128x48 pixels. We follow [12, 3] in considering Cam B as the gallery set and Cam A as the probe set. Performance is evaluated by matching each test image in Cam A against the Cam B gallery.

PRID is provided as both multi-shot and single-shot data. It consists of two camera views overlooking an urban environment from a distance and from fixed viewpoints. As a result PRID features low pose variability with the majority of people captured in profile. The first 200 shots in each view correspond to the same person, however the remaining shots only appear once in the dataset. To maximise comparability with VIPeR, we use the single-shot version and use the first 200 shots from each view. Images are scaled to 128x64 pixels.

For each dataset, we divide the available data into training, validation and test partitions. We initially train classifiers and produce attribute representations from the training portion, and then optimise the attribute weighting as described in Section 4.3.6 using the validation set. We then retrain the classifiers on both the training and validation portions, while re-identification performance is reported on the held out test portion.

We quantify re-identification performance using three standard metrics and one less common one metric. The standard re-identification metrics are performance at rank n , cumulative matching characteristic (CMC) curves, and normalised area under the CMC curve [12, 3]. Performance at rank n reports the probability that the correct match occurs within the first n ranked results from the gallery. The CMC curve plots this value for all n , and the nAUC summarises the area under the CMC curve (so perfect nAUC is 1.0 and chance nAUC is 0.5).

We additionally report Expected Rank (ER), as advocated by Avraham et al. [2] as CMC Expectation. The ER reflects the mean rank of the true matches and is a useful statistic for our purposes; in contrast to the standard metrics, lower ER scores are more desirable and indicate that on average the correct matches are distributed more toward the lower ranks. (So perfect ER is 1 and random ER would be half the gallery size). In particular ER has the advantage of a highly relevant practical interpretation: it is the average number of returned images the operator will have to scan before reaching the true match.

We compare the following re-identification methods: (1) SDALF [3] using code provided by the authors (note that SDALF is already shown to decisively outperform [13]); (2) ELF: Prosser et al.'s [37] spatial variant of Ensemble of Localised Features (ELF) [12] using Strips of ELF; (3) Attributes: Raw attribute based re-identification (Euclidean distance); (4) **OAR**: our Optimised Attribute based Re-identification

method with weighting between low-level features and within attributes learned by directly minimising the Expected Rank (Section 4.3.6).

4.4.2 Attribute Analysis

We first analyse the intrinsic discriminative potential of our attribute ontology independently of how reliably detectable the attributes are (assuming perfect detectability). This analysis provides an upper bound of performance that would be obtainable with sufficiently advanced attribute detectors. Fig. 4.6 reports the prevalence of each attribute in the datasets. Many attributes have prevalence near to 50%, which is reflected in their higher mutual information with person identity. As we discussed earlier this is a desirable property because it means each additional attribute known can potentially halve the number of possible matches. Whether this is realised or not depends on if attributes are correlated/redundant, in which case each additional redundant attribute provides less marginal benefit. To check this we compute the correlation coefficient between all attributes, and found that the average inter-attribute correlation was only 0.07. We therefore expect the attribute ontology to be effective.

Fig. 4.4 shows a histogram summarising how many people are uniquely identifiable solely by attributes and how many would be confused to a greater or lesser extent. The peak around unique/unambiguous shows that a clear majority of people can be uniquely or otherwise near-uniquely identified by their attribute-profile alone, while the tail shows that there are a small number of people with very generic profiles. This observation is important; near-uniqueness means that approaches which rank distances between attribute-profiles are still likely to feature the correct match high enough in the ranked list to be of use to human operators.

The CMC curve (for gallery size $p=632$) that would be obtained assuming perfect attribute classifiers is shown in Fig. 4.5. This impressive result (nAUC near a perfect score of 1.0) highlights the potential for attribute-based re-identification. Also shown are the results with only the top 5 or 10 attributes (sorted by mutual information with identity), and a random 10 attributes. This shows that: (i) as few as 10 attributes are sufficient if they are good (i.e. high MI) and perfectly detectable, while 5 is too few; and (ii) attributes with high MI are significantly more useful than low MI (always present or absent) attributes.

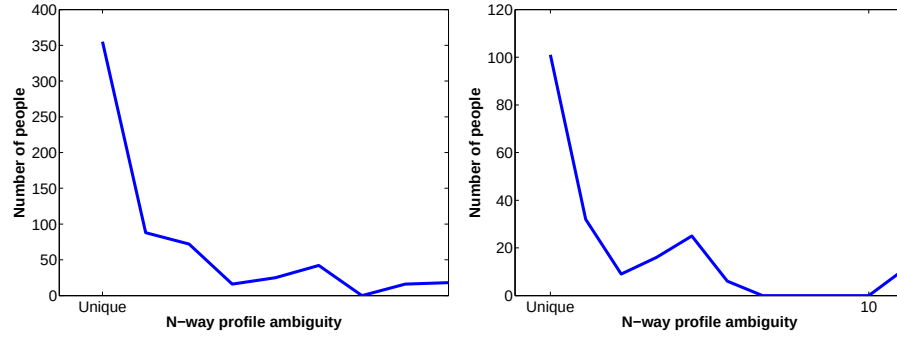


Fig. 4.4: Uniqueness of attribute descriptions in a population, (i) VIPeR and (ii) PRID. The peak around unique shows that most people are uniquely identifiable by attributes.

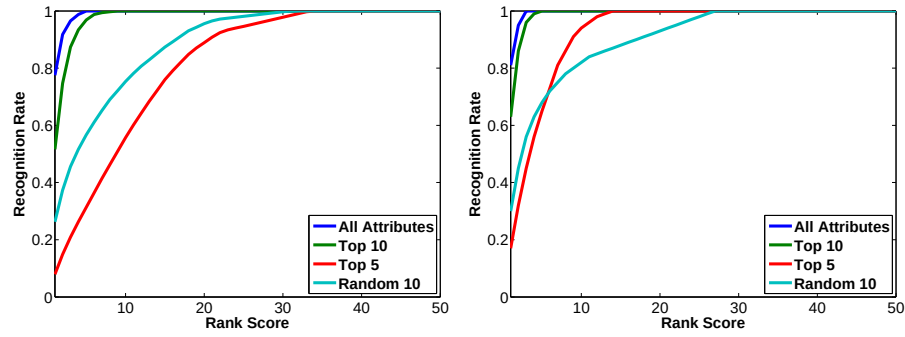


Fig. 4.5: Best-case (assuming perfect attribute detection) re-identification using attributes with highest n ground-truth Mutual Information scores, (i) VIPeR and (ii) PRID.

4.4.3 Attribute Detection

Given the analysis of the intrinsic effectiveness of the ontology in the previous section, the next question is whether the selected attributes can indeed be detected or not. Attribute detection on both VIPeR and PRID achieves reasonable levels on both balanced and unbalanced datasets as seen in Table 4.2. (dash indicates failure to train due to insufficient data). For all datasets, a minimum of 9 classifiers can be trained on unbalanced PRID, and 16 on unbalanced VIPeR, in both cases some attribute classifiers are unable to train due to extreme class imbalances or data sparsity. Average accuracies for these datasets are also reasonable; 66.9% and 68.3% respectively.

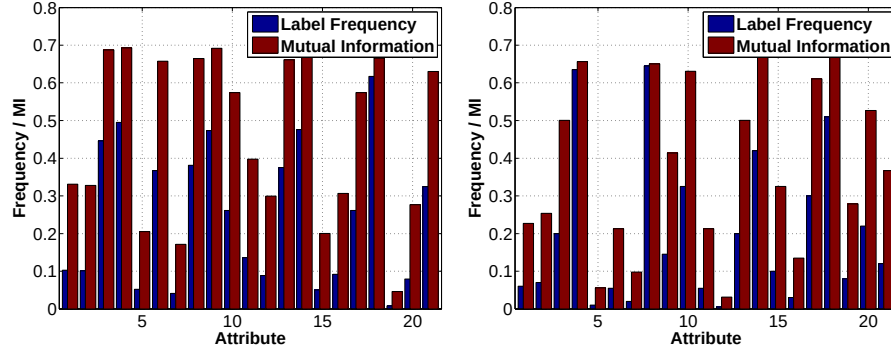


Fig. 4.6: Attribute occurrence frequencies and Attribute Mutual Information (MI) scores in VIPeR (left) and PRID (right).

Table 4.2: Attribute Classifier training and test accuracies (%) for VIPeR and PRID, for both the balanced (b) and unbalanced (ub) datasets.

	VIPeR (u)	VIPeR (b)	PRID (u)	PRID (b)
redshirt	79.6	80.9	–	41.3
blueshirt	62.7	68.3	–	59.6
lightshirt	80.6	82.2	81.6	80.6
darkshirt	82.2	84.0	79.0	79.5
greenshirt	57.3	72.1	–	–
nocoats	68.5	69.7	–	31.3
notlightdarkjeanscolour	57.6	69.1	–	–
darkbottoms	74.4	75.0	72.2	67.3
lightbottoms	75.3	74.7	76.0	74.0
hassatchel	–	56.0	51.9	55.0
barelegs	60.4	74.4	–	50.2
shorts	53.1	76.1	–	–
jeans	73.6	78.0	57.1	69.4
male	66.7	68.0	52.1	54.0
skirt	–	68.8	–	44.6
patterned	–	60.8	–	–
midhair	55.2	64.6	69.4	70.4
darkhair	60.0	60.0	75.4	75.4
bald	–	–	–	40.2
hashandbagcarrierbag	–	54.5	–	59.4
hasbackpack	63.4	68.6	–	48.3
Mean	66.9	70.3	68.3	66.2

The benefit of sub-sampling negative data for attribute learning is highlighted in the improvement in the balanced datasets. Balancing in this case increases the number of successfully trained classifiers to 20 for balanced VIPeR and 16 on balanced

Table 4.3: Re-identification performance, we report Expected Rank scores for VIPeR (left, gallery size $p = 316$) and PRID (right, gallery size $p = 100$) and compare different features and distance measures against our balanced attribute-features prior to fusion and weight selection. Smaller values indicate better re-identification performance.

VIPeR	$L1$	$L2$	PRID	$L1$	$L2$
ELF [37]	84.3	72.1	ELF	28.2	37.0
ELF PCA	85.3	74.5	ELF PCA	32.7	38.1
Raw Attributes	34.4	37.8	Raw Attributes	24.1	24.4
SDALF [3]	44.0		SDALF [3]	31.8	
Random Chance	158		Random Chance	50	

PRID with mean accuracies rising to 70.3% for VIPeR. Balancing slightly reduces classification performance on PRID to an average of 66.2%.

4.4.4 Using Attributes to Re-identify

Given the previous analysis of discriminability and detectability of the attributes, we now address the central question of attributes for re-identification. We first consider vanilla attribute re-identification (no weighting or fusion; $\mathbf{w}^L = 0, \mathbf{w}_a = 1$ in Eq. (4.1)). The re-identification performance of attributes alone is summarised in Table 4.3 in terms of expected rank. There are a few interesting points to note: (i) In most cases using $L2$ NN matching provides lower ER scores than $L1$ NN matching. (ii) On VIPeR and PRID, SDALF outperforms the other low-level features, and outperforms our basic attributes in VIPeR. (iii) Although the attribute-centric re-identification uses the *same low-level input features* (ELF), and the same $L1/L2$ NN matching strategy, attributes decisively outperform raw ELF. We can verify that this large difference is due to the semantic attribute space rather than the implicit dimensionality reduction effect of attributes by performing Principle Components Analysis (PCA) on ELF to reduce its dimensionality to the same as our attribute space ($N_a = 21$). In this case the re-identification performance is still significantly worse than the attribute-centric approach (See Table 4.3). The improvement over raw ELF is thus due to the attribute-centric approach.

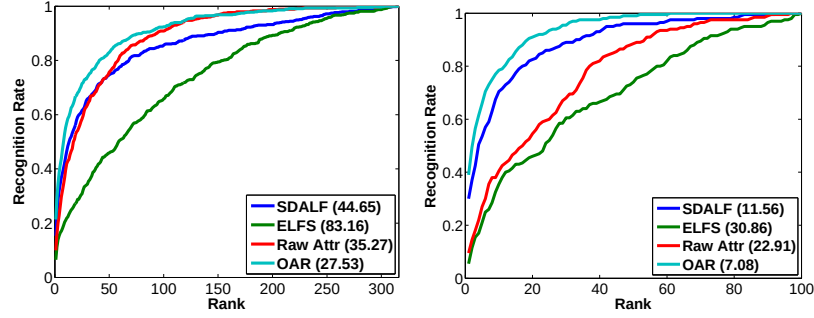


Fig. 4.7: Final attribute re-identification CMC plots for (i) VIPeR and (ii) PRID, Gallery sizes $p = 316, p = 100$. Expected Rank is given in parentheses.

Table 4.4: Final attribute re-identification performance. We report Expected Rank scores [2] (lower scores indicate that overall, an operator will find the correct match appears lower down the ranks), Cumulative Match Characteristic (CMC) and normalised Area-Under-Curve (nAUC) scores (higher is better, the maximum nAUC score is 1). We further report accuracies for our approach using unbalanced data for comparison.

VIPeR	ER	Rank 1	Rank 5	Rank10	Rank25	nAUC
Farenzena et al. [3]	44.7	15.3	34.5	44.3	61.6	0.86
Prosser et al. [37]	83.2	6.5	16.5	21.0	30.9	0.74
Raw Attributes (b)	35.3	10.0	26.3	39.6	58.4	0.89
OAR (b)	27.5	21.4	41.5	55.2	71.5	0.94
Raw Attributes (u)	40.4	6.5	23.9	34.8	55.9	0.88
OAR (u)	29.0	19.6	39.7	54.1	71.2	0.91
PRID	ER	Rank 1	Rank 5	Rank10	Rank25	nAUC
Farenzena et al.	11.6	30.0	53.5	70.5	86.0	0.89
Prosser et al.	30.9	5.5	21.0	35.5	52.0	0.70
Raw Attributes (b)	22.9	9.5	27.0	40.5	60.0	0.78
OAR (b)	7.1	39.0	66.0	78.5	93.5	0.93
Raw Attributes (u)	20.8	8.5	28.5	44.0	69.0	0.80
OAR (u)	6.2	41.5	69.0	82.5	95.0	0.95

Table 4.5: Comparison of results between our OAR method (Optimised Attribute Re-identification) and other state-of-art results for the VIPeR dataset.

VIPeR	Rank 1	Rank 10	Rank 20	Rank 50	nAUC
OAR	21.4	55.2	71.5	82.9	0.92
Hirzer et al.[16]	22.0	63.0	78.0	93.0	-
Farenzena et al.[3]	9.7	31.7	46.5	66.6	0.82
Hirzer et al.[17]	27.0	69.0	83.0	95.0	-
Avraham et al.[2]	15.9	59.7	78.3	-	-
Zheng et al.[47, 50]	15.7	53.9	70.1	-	-
Prosser et al.[37]	14.6	50.9	66.8	-	-

4.4.5 Re-Identification With Optimised Attributes

Given the promising results for vanilla attribute re-identification in the previous section, we finally investigate whether our complete model (including discriminative optimisation of weights to improve expected rank) can further improve performance. Fig. 4.7 and Table 4.4 summarise final re-identification performance. In each case, optimising the attributes with the distance metric and fusing with low-level SDALF and ELF improves re-identification uniformly compared to using attributes or low-level features alone. Our approach improves ER by 38.3% and 35% on VIPeR, and 38.8% and 46.5% on PRID for the balanced and unbalanced cases vs SDALF and 66.9%, 65.1%, 77.1% and 80% vs ELF features.

Critically for re-identification scenarios, the most important rank 1 accuracies are improved convincingly. For VIPeR, OAR improves 40% over SDALF in the balanced case, and 33.3% for unbalanced data. For PRID, OAR improves by 30% and 36.6%. As in the case of ER, rank is uniformly improved, indicating the increased likelihood that correct matches appear more frequently at earlier ranks using our approach.

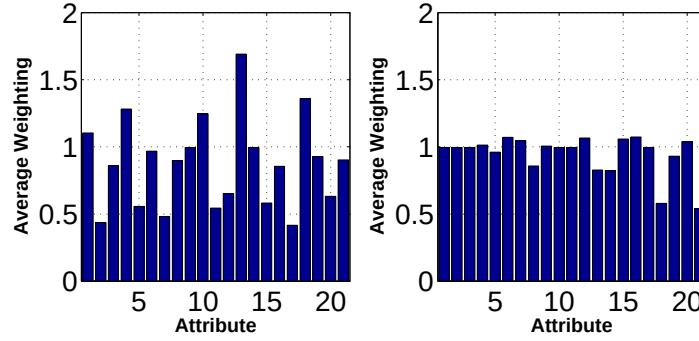


Fig. 4.8: Final attribute feature weights for VIPeR (left) and PRID (right).

The learned weights for fusion between our attributes and low-level features indicate that SDALF is informative and useful for re-identification on both datasets. In contrast, ELF is substantially down-weighted to 18% compared to SDALF on PRID and on VIPeR, disabled entirely. This makes sense because SDALF is at least twice as effective as ELF for VIPeR (Table 4.3).

The intra-attribute weights (Fig. 4.8) are relatively even on PRID but more varied on VIPeR where the highest weighted attributes (*jeans*, *hasbackpack*, *nocoats*, *midhair*, *shorts*) are weighted at 1.43, 1.20, 1.17, 1.10 and 1.1; while the least informative attributes are *barelegs*, *lightshirt*, *greenshirt*, *patterned* and *hassatchel* which are weighted to 0.7, 0.7, 0.66, 0.65 and 0.75. Jeans is one of the attributes that is detected most accurately and is most common in the datasets, so its weight is expected to be high. However the others are more surprising, with some of the most accurate

attributes such as *darkshirt* and *lightshirt* weighted relatively low (0.85 and 0.7). For PRID, *darkshirt*, *skirt*, *lightbottoms*, *lightshirt* and *darkbottoms* are most informative (1.19, 1.04, 1.02 and 1.03); *darkhair*, *midhair*, *bald*, *jeans* are the least (0.78, 0.8, 0.92, 0.86).

Interestingly, the most familiar indicators which might be expected to differentiate good versus bad attributes are not reflected in the final weighting. Classification accuracy, annotation error (label noise) and mutual information are not significantly correlated with the final weighting, meaning that some unreliably detectable and rare/low MI attributes actually turn out to be *useful* for re-identification with low expected rank; and vice-versa. Moreover, some of the weightings vary dramatically between dataset, for example, the attribute *jeans* is the strongest weighted attribute on VIPeR, however it is one of the lowest on PRID despite being reasonably accurate and prevalent on both datasets. These two observations both show (i) the necessity of jointly learning a combined weighting for all the attributes, (ii) doing so with a relevant objective function (such as ER), and (iii) learning a model which is adapted for the statistics of each given dataset/scenario.

In Table 4.5, we compare our approach with the performance other methods as reported in their evaluations. In this case the cross-validation folds are not the same, so the results are not exactly comparable, however they should be indicative. Our approach performs comparably to [16] and convincingly compared to [3, 47, 50] and [37]. Both [17] and [2] exploit pairwise learning; in [2] a binary classifier is trained on correct and incorrect pairs of detections in order to learn the projection from one camera to another, in [17] incorrect (i.e., matches that are nearer to the probe than the true match) detections are directly mapped further away whilst similar but correct matches are mapped closer together. Our approach is eventually outperformed by [17], however [17] learns a full covariance distance matrix in contrast to our simple diagonal matrix, and despite this we remain reasonably competitive.

4.4.6 Zero-shot Identification

In Section 4.4.2 we showed that with perfect attribute detections, highly accurate re-identification is possible. Even with a mere 10 attributes, near-perfect re-identification can be performed. Zero-shot identification is the task of generating an attribute-profile either manually or from a different modality of data, then matching individuals in the gallery set via their attributes. This is highly topical for surveillance: consider the case where a suspect is escaping through a public area surveilled by CCTV. The authorities in this situation may have enough information build a semantic-attribute-profile of the suspect using attributes taken from eyewitness descriptions.

In zero-shot identification (a special case of re-identification) we replace the probe image with a manually specified attribute description. To test this problem setting, we match the ground truth attribute-profiles of probe persons against their inferred attribute-profiles in the gallery as in [43].

An interesting question one might ask is whether this is expected to be better or worse than conventional attribute-space re-identification based on attributes detected from a probe *image*. One might expect zero-shot performance to be better because we know that in the absence of noise, attribute re-identification performs admirably (Section 4.4.2 and Fig. 4.5) – and there are two sources of noise (attribute detection inaccuracies in the probe and target images) of which the former noise source has been removed in the zero-shot case. In this case, a man-in-the-loop approach to querying might be desirable, even if a probe image is available. That is, the operator could quickly indicate the ground-truth attributes for the probe image and search based on this (noise-free) ground-truth.

Table 4.6 shows re-identification performance for both datasets. Surprisingly, while the performance is encouraging, it is not as compelling as when the profile is constructed by our classifiers, *despite the elimination of the noise on the probe images*.

This significant difference between the zero-shot case we outline here and the conventional case we discuss in the previous section turns out to be because of *noise correlation*. Intuitively, consider that if someone with a hard-to-classify hairstyle is classified in one camera with some error ($p(a_{\text{hair}}|\mathbf{x}) - a_{\text{hair}}^{\text{true}}$), then this person might also be classified in another camera with an error *in the same direction*. In this case, using the ground-truth attribute in one camera will actually be detrimental to re-identification performance.

To verify this explanation, we perform Pearson’s product-moment correlation analysis on the error (difference between ground-truth labels and the predicted attributes) between the probe and gallery sets. The average cross-camera error correlation coefficient is 0.93 in VIPeR and 0.97 in PRID, and all of the correlation coefficients were statistically significant ($p < 0.05$).

Although these results show that man-in-the-loop zero-shot identification - if intended to replace a probe image - may not always be beneficial, it is still evident that zero-shot performs reasonably in general and is a valuable capability for the case where descriptions are verbal rather than extracted from a visual example.

Table 4.6: Zero-shot re-identification results for VIPeR and PRID.

	ExpRank	Rank 1	Rank 5	Rank10	Rank25
VIPeR (u)	50.1	6.0	17.1	26.0	48.1
VIPeR (b)	54.8	5.4	14.9	25.3	44.9
PRID (u)	19.2	8.0	29.0	47.0	73.0
PRID (b)	26.1	3.0	16.0	32.0	62.0

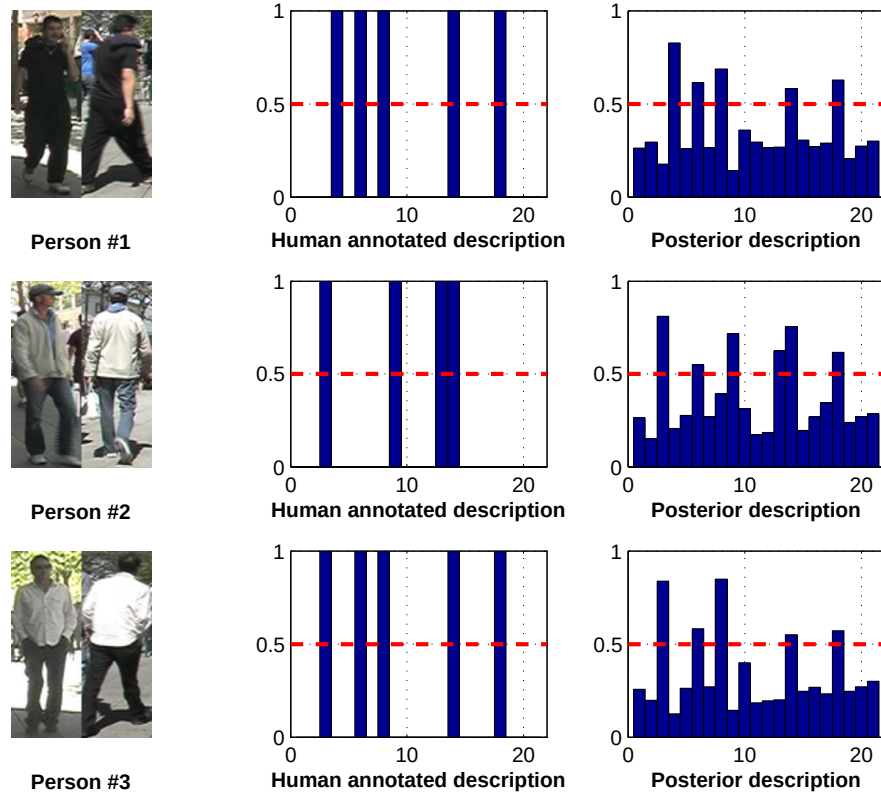


Fig. 4.9: Success cases for Zero-shot re-identification on VIPeR. The left column shows two probe images; i) is the image annotated by a human operator and ii) is the correct rank #1 match as selected by our zero-shot re-identification system. The human-annotated probe descriptions (middle) and the matched attribute-feature gallery descriptions (right) are notably similar for each person; the attribute detections from the gallery closely resemble the human-annotated attributes (particularly those above red line).

4.5 Conclusions

We have shown how mid-level attributes are trained using semantic cues from human experts [33] can be an effective representation for re-identification and (zero-shot) identification. Moreover, this provides a different modality to standard low-level features and thus synergistic opportunities for fusion.

Existing approaches to re-identification [3, 37, 12] focus on high-dimensional low-level features which aim to be discriminative for identity yet invariant to view and lighting. However, these variance and invariance properties are hard to obtain si-

multaneously, thus limiting such features effectiveness for re-identification. In contrast, attributes provide a low-dimensional mid-level representation which are discriminative by construction (see Section 4.3.1) and make no strong view invariance assumptions (variability in appearance of each attribute is learned by the classifier with sufficient training data).

Importantly, although individual attributes vary in robustness and informativeness, attributes provide a strong cue for identity. Their low-dimensional nature means they are also amenable to discriminatively learning a good distance metric, in contrast to the challenging optimisation required for high-dimensional LLFs [47, 50]. In developing a separate cue-modality, our approach is potentially complementary to the majority of existing approaches, whether focused on low-level features [3], or learning methods [47, 50].

The most promising direction for future research is improving the attribute-detector performance, as evidenced by the excellent results in Fig. 4.5 using ground-truth attributes. The more limited empirical performance is due to lack of training data, which could be addressed by transfer learning to deploy attribute detectors trained on large databases (e.g. web-crawls) on to the re-identification system.

4.6 Further Reading

Interested readers may wish to refer to the following material:

- [32] for a comprehensive overview of continuous optimisation methods.
- [31] for detailed exposition and review of contemporary features and descriptors.
- [30] discusses classifier training and machine learning methods.
- [39] for trends on surveillance hardware development.

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